

Metaphysical Foundations of the Multi-Representation Neurosymbolic Forest

System Architecture Invariant Manifest

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1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta’s Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past (e.g., Moses, David). They are structurally locked against retroactive modifications.
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites (e.g., John, Judith, Pierre).
3. **The Indexical Real World:** The concrete evaluation layer where sovereign actions and executions occur in real time.

2 The Forest Layout: Three Independent Structural Representations

To prevent feature-space pollution and stabilize Shannon Entropy boundaries during statistical induction, the architecture splits processing tasks across three decoupled upper-tier decision models.

2.1 Representation I: Causal Counterfactuals (Tree #5)

This model monitors binary concept assertions ($\text{Sign} \in \{+, -\}$) to maintain narrative coherence under unexpected contextual breaks. The system

evaluates the state profiles using the split rule:

$$\mathcal{F}(s') = \begin{cases} \text{preserve_semantics_equivalence,} & \text{if Sign} = + \\ \text{trigger_terminal_node_rewrite,} & \text{if Sign} = - \end{cases} \quad (1)$$

2.1.1 Exemplar Trace: Causal Job-Loss Counterfactual (ST1)

Let S_{ST1} be a multi-sentence situation tracking a worker’s trajectory. The transition from positive baseline diligence to an active withdrawal state is formulated strictly as follows:

$$\begin{aligned} \text{ST1}_{\text{pos}} &= \langle \text{stagnate, diligent, rejection, withdraw} \rangle \longrightarrow \text{quits_job} \\ \text{ST1}_{\text{neg}} &= \langle \text{stagnate, diligent, } \neg \text{rejection, } \neg \text{withdraw} \rangle \longrightarrow \text{keeps_job} \end{aligned}$$

When the request for a raise is refused ($\text{Sign} = -$), Tree #5 triggers a structural branch rewrite, allowing the agent to exit the active edge safely rather than falling into an unhandled system failure.

2.2 Representation II: FCA Parthood Ontologies (Tree #6)

Using Formal Concept Analysis (FCA), situations are decomposed into an 8-dimensional attribute matrix mapping abstract structural parthood ($\sqsubseteq_{\text{part_of}}$). The categories are defined across uniform relational axes:

$$R_{\text{FCA}} = \langle \text{Syntax, Type, Region, Event, Argument, Action, Set, Time} \rangle \quad (2)$$

This matrix isolates multi-sentence overlaps natively. For example, it maps the geographical context of *Egypt* symmetrically across both Moses and Joseph, while cleanly splitting their terminal classifications based on distinct argument vectors (e.g., *killing_egyptian* vs. *youthful_pride*).

2.3 Representation III: Relational Role Homomorphisms (Tree #7)

Analogies are not evaluated as loose word overlaps. They are treated as distinct complex abstract objects defined by a structural partial homomorphism ($h : S \rightarrow T$) from a Source domain to a Target domain.

2.3.1 Pragmatic Social Contract Constraints

Tree #7 checks for high-order pragmatic governance rules. Given an action p and a mandatory permission requirement q , an unauthorized exception

state triggers an immediate contract breach vertex:

$$\text{Pragmatic_Breach}(p, q) \iff p \wedge \neg q \quad (3)$$

For instance, a house kitchen fire expanding while an emergency dispatcher is distracted directly maps onto a mountain emergency where the baseline witness (Miranda) is asleep ($p \wedge \neg q$).

3 Analogical Problem Solving via Graph Minors

When a target journey line encounters an unexpected topological link break, the system utilizes graph-minor analogical inference to compute a path repair.

Let a planned trajectory be represented as a base network graph $G = (V, E)$.

1. **Edge Deletion ($G \setminus e$):** Disruption events (e.g., John missing his morning Amtrak train due to a track obstruction) delete a critical transit link $e \in E$, rendering the destination V_{goal} unreachable.
2. **Node Contraction (G/e):** The recovery strategy acts as a node contraction operator. By calling a local cab, the system dynamically folds the broken pathway coordinates back together, restructuring the topological space to preserve destination connectivity.

The accumulated non-linear transformation cost is calculated natively over the Grigorchuk Infinite Group Matrix by traversing equivalent vertex steps inside the background knowledge laws:

$$\text{TotalCost} = \sum \text{EqCost} + \text{StepCost} + \text{SubCost} \quad (4)$$

4 The Spiritual Analogy Component: Judith and the Prodigal Son

The highest layer of analogical inference ascends completely out of the mechanical infrastructure to evaluate structural life-track recovery loops. We define a high-level relational role homomorphism mapping Judith's mountain descent to the narrative of the Prodigal Son ($h : \text{Judith} \rightarrow \text{Prodigal_Son}$):

Source Scenario (Judith on Mt. Ateh)	$\xrightarrow{h}$	Target Scenario (The Prodigal Son)
Ascent under ease and abundance	$\longrightarrow$	Initial inheritance and comfortable safety
Decision to take a different unverified route	$\longrightarrow$	Departure to a far, unfamiliar country
Helpful piles of stones completely peter out	$\longrightarrow$	The onset of severe famine and exhaustion
Hopelessly lost and fallen on a scree slope	$\longrightarrow$	Stranded in poverty, feeding swine in the field
Injured and trapped against a hardy thorn bush	$\longrightarrow$	The awakening event: coming to his senses
Remembering and flashing the working flashlight	$\longrightarrow$	The return journey and confession: "I have sinned"
Helicopter searchlight illuminates the face	$\longrightarrow$	The Father running out to embrace and rescue
Waving joyously at the incoming aircraft	$\longrightarrow$	The ultimate feast of celebration and redemption

4.1 The Metaphysical Conclusion

The flashlight beacon and the son's confession serve identical roles as **Informational Support Links**. They transform a lost coordinate position into an active, tethered rescue destination. Because both stories map to the exact same high-level mathematical lattice structure inside Tree #7, the system proves that the transition from a broken fall to a sovereign rescue is governed by an invariant top-down law of redemption.